CONSOLIDATED RESEARCH LITERATURE REVIEW

Domain-Adaptive and Robustness-Aware Question Answering:
A Comparative Study of Fine-Tuned BERT Variants Under Data-Scarce and Adversarial Conditions

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45 unique papers • 22 foundational papers plus additional domain, data-scarcity and robustness research

Includes 2025–2026 near-neighbour studies and a project-specific research-gap analysis.

Scope, verification and interpretation notes

Bibliographic verification: Titles, author lists, publication venues/years and paper links were checked against primary publisher, ACL Anthology, arXiv, IEEE, ACS, PubMed Central or other official records where available. arXiv-only work is labelled as a preprint.

Content verification: Summaries use official abstracts, publisher records and accessible full-text sections. They are analytical syntheses, not verbatim reproductions of original paper sections. For final submission, inspect linked full texts for exact experimental settings, tables, hyperparameters and detailed numerical claims.

Eight-section format: Abstract, Introduction, Related Work, Methodology, Experiments, Results, Limitations and Conclusion are structured summaries. “Related Work” positions each paper within this review's corpus; it does not reproduce the paper's own Related Work section.

Task boundary: The proposed task is extractive technical-domain QA: a model selects an answer span from a supplied passage. Open-domain RAG, multilingual, spoken and multi-hop QA papers are supporting or adjacent work, not direct equivalents.

Deduplication: Duplicate .webloc links from the original collection and repeated appearances across the recent-research/data-scarcity collections are counted once.

Collection overview

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1. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Authors: Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova
Publication: NAACL-HLT, 2019
Research area: Foundations & model comparison
Primary source: https://arxiv.org/abs/1810.04805

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Bidirectional Transformer pretraining with masked language modelling and next-sentence prediction, followed by task-specific fine-tuning. Evidence/result reported in the paper: Establishes a strong general-purpose pretrained encoder and a widely used extractive-QA baseline.
2	Introduction	The central problem is large pretrained models require compute. The paper studies this within foundations & model comparison.
3	Related Work	This work is most closely connected to foundations & model comparison and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Bidirectional Transformer pretraining with masked language modelling and next-sentence prediction, followed by task-specific fine-tuning.
5	Experiments	Evaluates language understanding benchmarks including SQuAD; extractive QA predicts start and end answer positions.
6	Results	Establishes a strong general-purpose pretrained encoder and a widely used extractive-QA baseline.
7	Limitations	Large pretrained models require compute; performance depends on fine-tuning and domain match.
8	Conclusion	For this project, the defensible takeaway is: Use BERT-base as the baseline encoder and common span-prediction formulation. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use BERT-base as the baseline encoder and common span-prediction formulation.

2. RoBERTa: A Robustly Optimized BERT Pretraining Approach

Authors: Yinhan Liu et al.
Publication: arXiv preprint, 2019
Research area: Foundations & model comparison
Primary source: https://arxiv.org/abs/1907.11692

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Optimizes BERT pretraining with more data, longer training, larger batches, dynamic masking and removal of NSP. Evidence/result reported in the paper: Improved pretraining choices produce stronger benchmark results than the original BERT recipe.
2	Introduction	The central problem is compute and pretraining scale complicate direct architecture-only comparisons. The paper studies this within foundations & model comparison.
3	Related Work	This work is most closely connected to foundations & model comparison and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Optimizes BERT pretraining with more data, longer training, larger batches, dynamic masking and removal of NSP.
5	Experiments	Evaluates GLUE, SQuAD and other language-understanding benchmarks.
6	Results	Improved pretraining choices produce stronger benchmark results than the original BERT recipe.
7	Limitations	Compute and pretraining scale complicate direct architecture-only comparisons.
8	Conclusion	For this project, the defensible takeaway is: Compare RoBERTa with BERT under identical downstream data and tuning budgets. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Compare RoBERTa with BERT under identical downstream data and tuning budgets.

3. DeBERTa: Decoding-enhanced BERT with Disentangled Attention

Authors: Pengcheng He, Xiaodong Liu, Jianfeng Gao, Weizhu Chen
Publication: ICLR, 2021
Research area: Foundations & model comparison
Primary source: https://arxiv.org/abs/2006.03654

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Uses disentangled content/position attention and an enhanced mask decoder. Evidence/result reported in the paper: Reports improvements over comparable BERT-family baselines on several benchmarks.
2	Introduction	The central problem is additional architectural complexity and compute may affect fair comparisons. The paper studies this within foundations & model comparison.
3	Related Work	This work is most closely connected to foundations & model comparison and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Uses disentangled content/position attention and an enhanced mask decoder.
5	Experiments	Evaluates GLUE, SQuAD and other language-understanding benchmarks.
6	Results	Reports improvements over comparable BERT-family baselines on several benchmarks.
7	Limitations	Additional architectural complexity and compute may affect fair comparisons.
8	Conclusion	For this project, the defensible takeaway is: Test whether disentangled attention improves technical QA and robustness. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Test whether disentangled attention improves technical QA and robustness.

4. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter

Authors: Victor Sanh, Lysandre Debut, Julien Chaumond, Thomas Wolf
Publication: NeurIPS EMC2 Workshop, 2019
Research area: Foundations & model comparison
Primary source: https://arxiv.org/abs/1910.01108

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Distills BERT into a smaller student model using language-model, distillation and cosine-distance objectives. Evidence/result reported in the paper: Retains much of BERT's performance with fewer parameters and faster inference.
2	Introduction	The central problem is compression may reduce capacity. The paper studies this within foundations & model comparison.
3	Related Work	This work is most closely connected to foundations & model comparison and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Distills BERT into a smaller student model using language-model, distillation and cosine-distance objectives.
5	Experiments	Evaluates GLUE and SQuAD, including speed and model-size comparisons.
6	Results	Retains much of BERT's performance with fewer parameters and faster inference.
7	Limitations	Compression may reduce capacity; efficiency does not guarantee robustness.
8	Conclusion	For this project, the defensible takeaway is: Include an efficient baseline and measure accuracy, latency and memory. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Include an efficient baseline and measure accuracy, latency and memory.

5. ALBERT: A Lite BERT for Self-supervised Learning of Language Representations

Authors: Zhenzhong Lan et al.
Publication: ICLR, 2020
Research area: Foundations & model comparison
Primary source: https://arxiv.org/abs/1909.11942

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Uses factorized embedding parameterization, cross-layer parameter sharing and sentence-order prediction. Evidence/result reported in the paper: Achieves competitive language-understanding results with fewer parameters.
2	Introduction	The central problem is parameter efficiency does not automatically imply domain adaptation or adversarial robustness. The paper studies this within foundations & model comparison.

No.	Section	Structured analysis
3	Related Work	This work is most closely connected to foundations & model comparison and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Uses factorized embedding parameterization, cross-layer parameter sharing and sentence-order prediction.
5	Experiments	Evaluates GLUE, SQuAD and RACE.
6	Results	Achieves competitive language-understanding results with fewer parameters.
7	Limitations	Parameter efficiency does not automatically imply domain adaptation or adversarial robustness.
8	Conclusion	For this project, the defensible takeaway is: Compare parameter-efficient architecture performance and compute cost. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Compare parameter-efficient architecture performance and compute cost.

6. SQuAD: 100,000+ Questions for Machine Comprehension of Text

Authors: Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, Percy Liang
Publication: EMNLP, 2016
Research area: Foundations & datasets
Primary source: https://arxiv.org/abs/1606.05250

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Crowdworkers write questions on Wikipedia passages and identify answer spans. Evidence/result reported in the paper: Provides a standard benchmark for extractive QA.
2	Introduction	The central problem is wikipedia is general-domain. The paper studies this within foundations & datasets.
3	Related Work	This work is most closely connected to foundations & datasets and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Crowdworkers write questions on Wikipedia passages and identify answer spans.
5	Experiments	Over 100,000 question-answer pairs; commonly evaluated using Exact Match and token F1.
6	Results	Provides a standard benchmark for extractive QA.
7	Limitations	Wikipedia is general-domain; SQuAD 1.1 is answerable-only and does not represent technical terminology.
8	Conclusion	For this project, the defensible takeaway is: Use as a general-domain source baseline; add a separate technical-domain evaluation set. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use as a general-domain source baseline; add a separate technical-domain evaluation set.

7. Don't Stop Pretraining: Adapt Language Models to Domains and Tasks

Authors: Suchin Gururangan et al.
Publication: ACL, 2020
Research area: Domain adaptation
Primary source: https://arxiv.org/abs/2004.10964

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces Domain-Adaptive Pretraining (DAPT) and Task-Adaptive Pretraining (TAPT) through continued MLM training. Evidence/result reported in the paper: Continued pretraining can improve downstream performance when source and target text differ.
2	Introduction	The central problem is needs relevant unlabeled target-domain text and extra compute. The paper studies this within domain adaptation.
3	Related Work	This work is most closely connected to domain adaptation and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces Domain-Adaptive Pretraining (DAPT) and Task-Adaptive Pretraining (TAPT) through continued MLM training.
5	Experiments	Evaluates across domains and downstream NLP tasks.
6	Results	Continued pretraining can improve downstream performance when source and target text differ.

No.	Section	Structured analysis
7	Limitations	Needs relevant unlabeled target-domain text and extra compute; gains vary by domain.
8	Conclusion	For this project, the defensible takeaway is: Continue MLM pretraining on technical documents before QA fine-tuning. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Continue MLM pretraining on technical documents before QA fine-tuning.

8. BioBERT: a pre-trained biomedical language representation model

Authors: Jinhyuk Lee et al.
Publication: Bioinformatics, 2020
Research area: Domain adaptation
Primary source: https://arxiv.org/abs/1901.08746

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Continues BERT pretraining on PubMed abstracts and PMC full-text biomedical articles. Evidence/result reported in the paper: Biomedical-domain pretraining improves several biomedical NLP tasks.
2	Introduction	The central problem is biomedical findings do not automatically transfer to technical documentation. The paper studies this within domain adaptation.
3	Related Work	This work is most closely connected to domain adaptation and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Continues BERT pretraining on PubMed abstracts and PMC full-text biomedical articles.
5	Experiments	Evaluates biomedical NER, relation extraction and QA tasks.
6	Results	Biomedical-domain pretraining improves several biomedical NLP tasks.
7	Limitations	Biomedical findings do not automatically transfer to technical documentation.
8	Conclusion	For this project, the defensible takeaway is: Use as a domain-specific pretraining precedent, not as a ready-made technical model. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use as a domain-specific pretraining precedent, not as a ready-made technical model.

9. MRQA 2019 Shared Task: Evaluating Generalization in Reading Comprehension

Authors: Adam Fisch, Alon Talmor, Robin Jia, Minjoon Seo, Eunsol Choi, Danqi Chen
Publication: MRQA Workshop, 2019
Research area: Domain adaptation & evaluation
Primary source: https://arxiv.org/abs/1910.09753

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Standardizes multiple reading-comprehension datasets and evaluates cross-dataset generalization. Evidence/result reported in the paper: Shows that in-domain QA performance does not guarantee out-of-domain generalization.
2	Introduction	The central problem is dataset formats and annotation conventions differ. The paper studies this within domain adaptation & evaluation.
3	Related Work	This work is most closely connected to domain adaptation & evaluation and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Standardizes multiple reading-comprehension datasets and evaluates cross-dataset generalization.
5	Experiments	Uses multiple extractive QA datasets with held-out target datasets.
6	Results	Shows that in-domain QA performance does not guarantee out-of-domain generalization.
7	Limitations	Dataset formats and annotation conventions differ; technical manuals may be underrepresented.
8	Conclusion	For this project, the defensible takeaway is: Use held-out technical-domain testing and report in-/out-of-domain performance separately. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use held-out technical-domain testing and report in-/out-of-domain performance separately.

10. End-to-End Synthetic Data Generation for Domain Adaptation of Question Answering Systems

Authors: Siamak Shakeri, Cicero Nogueira dos Santos, Henghui Zhu, Patrick Ng, Feng Nan, Zhiguo Wang, Ramesh Nallapati, Bing Xiang
Publication: EMNLP, 2020
Research area: Domain adaptation & augmentation
Primary source: https://aclanthology.org/2020.emnlp-main.439/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: A single Transformer encoder-decoder generates a question and answer from a passage; generation likelihood serves as a filtering score. Evidence/result reported in the paper: Synthetic target-domain data can help adapt QA systems in the studied settings.
2	Introduction	The central problem is generated questions/answers may be noisy or unlike real technical queries. The paper studies this within domain adaptation & augmentation.
3	Related Work	This work is most closely connected to domain adaptation & augmentation and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	A single Transformer encoder-decoder generates a question and answer from a passage; generation likelihood serves as a filtering score.
5	Experiments	Evaluates generated QA data for domain adaptation on QA datasets; publisher abstract reports significant improvements over prior methods.
6	Results	Synthetic target-domain data can help adapt QA systems in the studied settings.
7	Limitations	Generated questions/answers may be noisy or unlike real technical queries.
8	Conclusion	For this project, the defensible takeaway is: Generate and validate technical QA pairs as an augmentation option. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Generate and validate technical QA pairs as an augmentation option.

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11. COVID-QA: A Question Answering Dataset for COVID-19

Authors: Patrick Möller, Daniel Reina, Andreas Jayakumar, Johannes Pietsch
Publication: NLP-COVID19 Workshop, 2020
Research area: Domain adaptation & datasets
Primary source: https://aclanthology.org/2020.nlpcovid19-acl.18/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Collects question-answer pairs and answer spans from COVID-19 scientific articles. Evidence/result reported in the paper: Provides a domain-specific QA benchmark and demonstrates specialization needs.
2	Introduction	The central problem is biomedical corpus characteristics differ from software and technical documentation. The paper studies this within domain adaptation & datasets.
3	Related Work	This work is most closely connected to domain adaptation & datasets and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Collects question-answer pairs and answer spans from COVID-19 scientific articles.
5	Experiments	Evaluates QA models on a specialized biomedical corpus.
6	Results	Provides a domain-specific QA benchmark and demonstrates specialization needs.
7	Limitations	Biomedical corpus characteristics differ from software and technical documentation.
8	Conclusion	For this project, the defensible takeaway is: Use as an example of specialized-domain QA; not as the target domain. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use as an example of specialized-domain QA; not as the target domain.

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12. Few-Shot Question Answering by Pretraining Span Selection

Authors: Ori Ram, Yuval Kirstain, Jonathan Berant, Amir Globerson, Omer Levy
Publication: ACL, 2021
Research area: Data scarcity
Primary source: https://arxiv.org/abs/2101.00438

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces Recurring Span Selection pretraining to improve few-shot extractive QA. Evidence/result reported in the paper: Task-relevant span-selection pretraining improves few-shot QA in reported settings.
2	Introduction	The central problem is adds a specialized pretraining objective and does not compare all planned architectures under three conditions. The paper studies this within data scarcity.
3	Related Work	This work is most closely connected to data scarcity and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces Recurring Span Selection pretraining to improve few-shot extractive QA.
5	Experiments	Evaluates few-shot QA across datasets and sample sizes.
6	Results	Task-relevant span-selection pretraining improves few-shot QA in reported settings.
7	Limitations	Adds a specialized pretraining objective and does not compare all planned architectures under three conditions.
8	Conclusion	For this project, the defensible takeaway is: Include low-shot learning curves and consider a few-shot-aware baseline. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Include low-shot learning curves and consider a few-shot-aware baseline.

13. On the Stability of Fine-tuning BERT: Misconceptions, Explanations, and Strong Baselines

Authors: Alexander Mosbach, Maksym Andriushchenko, Dietrich Klakow
Publication: ICLR, 2021
Research area: Data scarcity & reproducibility
Primary source: https://arxiv.org/abs/2006.04884

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Analyzes optimization dynamics, hyperparameters and random-seed effects in BERT fine-tuning. Evidence/result reported in the paper: Shows fine-tuning can be more stable with suitable settings; seed variance should be reported.
2	Introduction	The central problem is main experiments are not technical extractive qa and do not compare all proposed architectures. The paper studies this within data scarcity & reproducibility.
3	Related Work	This work is most closely connected to data scarcity & reproducibility and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Analyzes optimization dynamics, hyperparameters and random-seed effects in BERT fine-tuning.
5	Experiments	Controlled experiments on GLUE tasks and different fine-tuning settings.
6	Results	Shows fine-tuning can be more stable with suitable settings; seed variance should be reported.
7	Limitations	Main experiments are not technical extractive QA and do not compare all proposed architectures.
8	Conclusion	For this project, the defensible takeaway is: Run repeated seeds and report mean and standard deviation. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Run repeated seeds and report mean and standard deviation.

14. Revisiting Few-sample BERT Fine-tuning

Authors: Tianyi Zhang, Felix Wu, Arzoo Katiyar, Kilian Q. Weinberger, Yoav Artzi
Publication: ICLR, 2021
Research area: Data scarcity & reproducibility
Primary source: https://arxiv.org/abs/2006.05987

Source note: Check complete author names and proceedings record against the original source before formal citation.

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Studies fine-tuning configurations for BERT with very small labeled datasets. Evidence/result reported in the paper: Identifies sources of instability and finds that some proposed few-shot methods have diminished impact under the revised training procedure.
2	Introduction	The central problem is not specifically technical qa or adversarial robustness. The paper studies this within data scarcity & reproducibility.

No.	Section	Structured analysis
3	Related Work	This work is most closely connected to data scarcity & reproducibility and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Studies fine-tuning configurations for BERT with very small labeled datasets.
5	Experiments	Studies few-sample BERT fine-tuning instabilities and tests alternative optimization/training practices.
6	Results	Identifies sources of instability and finds that some proposed few-shot methods have diminished impact under the revised training procedure.
7	Limitations	Not specifically technical QA or adversarial robustness; author metadata should be checked against source.
8	Conclusion	For this project, the defensible takeaway is: Use as a reference for low-data optimizer, layer and epoch choices. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use as a reference for low-data optimizer, layer and epoch choices.

15. Adversarial Examples for Evaluating Reading Comprehension Systems

Authors: Robin Jia, Percy Liang
Publication: EMNLP, 2017
Research area: Adversarial robustness
Primary source: https://arxiv.org/abs/1707.07328

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces AddSent and AddOneSent distractor-sentence attacks. Evidence/result reported in the paper: Shows that high benchmark scores can hide vulnerability to carefully inserted distractors.
2	Introduction	The central problem is attack family is limited to its designed setting and does not cover all modern attacks. The paper studies this within adversarial robustness.
3	Related Work	This work is most closely connected to adversarial robustness and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces AddSent and AddOneSent distractor-sentence attacks.
5	Experiments	Tests reading-comprehension systems on original and adversarially modified passages.
6	Results	Shows that high benchmark scores can hide vulnerability to carefully inserted distractors.
7	Limitations	Attack family is limited to its designed setting and does not cover all modern attacks.
8	Conclusion	For this project, the defensible takeaway is: Adapt distractor insertion and measure EM/F1 degradation. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Adapt distractor insertion and measure EM/F1 degradation.

16. Beat the AI: Investigating Adversarial Human Annotation for Reading Comprehension

Authors: Max Bartolo, Alastair Roberts, Johannes Welbl, Sebastian Riedel, Pontus Stenetorp
Publication: TACL, 2020
Research area: Adversarial robustness & datasets
Primary source: https://arxiv.org/abs/2002.00293

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces model-in-the-loop human adversarial question writing and the AdversarialQA dataset. Evidence/result reported in the paper: Adversarially written questions expose failures hidden by conventional benchmarks.
2	Introduction	The central problem is human annotation is costly and attack distribution depends on models and annotators. The paper studies this within adversarial robustness & datasets.
3	Related Work	This work is most closely connected to adversarial robustness & datasets and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces model-in-the-loop human adversarial question writing and the AdversarialQA dataset.
5	Experiments	Evaluates reading-comprehension systems on adversarially collected questions.
6	Results	Adversarially written questions expose failures hidden by conventional benchmarks.

No.	Section	Structured analysis
7	Limitations	Human annotation is costly and attack distribution depends on models and annotators.
8	Conclusion	For this project, the defensible takeaway is: Use AdversarialQA as an external robustness test set if formats are compatible. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use AdversarialQA as an external robustness test set if formats are compatible.

17. Is BERT Really Robust? A Strong Baseline for Natural Language Attack on Text Classification and Entailment

Authors: Di Jin, Zhijing Jin, Joey Tianyi Zhou, Peter Szolovits
Publication: AAAI, 2020
Research area: Adversarial robustness
Primary source: https://arxiv.org/abs/1907.11932

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces TextFooler, ranking important words and replacing them with semantically similar alternatives. Evidence/result reported in the paper: Demonstrates that meaning-preserving substitutions can fool NLP models.
2	Introduction	The central problem is not originally an extractive qa paper. The paper studies this within adversarial robustness.
3	Related Work	This work is most closely connected to adversarial robustness and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces TextFooler, ranking important words and replacing them with semantically similar alternatives.
5	Experiments	Original experiments focus on text classification and textual entailment.
6	Results	Demonstrates that meaning-preserving substitutions can fool NLP models.
7	Limitations	Not originally an extractive QA paper; attack adaptation must be validated.
8	Conclusion	For this project, the defensible takeaway is: Adapt lexical substitutions to QA questions/contexts and check answer-span consistency. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Adapt lexical substitutions to QA questions/contexts and check answer-span consistency.

18. FreeLB: Enhanced Adversarial Training for Natural Language Understanding

Authors: Chen Zhu, Yu Cheng, Zhe Gan, Siqi Sun, Tom Goldstein, Jingjing Liu
Publication: ICLR, 2020
Research area: Adversarial robustness & training
Primary source: https://arxiv.org/abs/1909.11764

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Applies multi-step adversarial perturbations in embedding space and accumulates gradients. Evidence/result reported in the paper: Reports gains on several NLU tasks using adversarial training.
2	Introduction	The central problem is not exclusively technical extractive qa. The paper studies this within adversarial robustness & training.
3	Related Work	This work is most closely connected to adversarial robustness & training and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Applies multi-step adversarial perturbations in embedding space and accumulates gradients.
5	Experiments	Evaluates on natural-language understanding benchmarks.
6	Results	Reports gains on several NLU tasks using adversarial training.
7	Limitations	Not exclusively technical extractive QA; training cost and task transfer need testing.
8	Conclusion	For this project, the defensible takeaway is: Test FreeLB-style fine-tuning as an optional robustness intervention. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Test FreeLB-style fine-tuning as an optional robustness intervention.

19. Robust Machine Comprehension Models via Adversarial Training

Authors: Yicheng Wang, Mohit Bansal
Publication: NAACL-HLT, 2018
Research area: Adversarial robustness
Primary source: https://arxiv.org/abs/1804.06473

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces AddSentDiverse to create more varied adversarial training examples and adds semantic-relation learning. Evidence/result reported in the paper: Reports a 36.5% F1 increase under several adversarial evaluations while maintaining regular SQuAD performance.
2	Introduction	The central problem is benefits depend on attack type and generated-example quality. The paper studies this within adversarial robustness.
3	Related Work	This work is most closely connected to adversarial robustness and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces AddSentDiverse to create more varied adversarial training examples and adds semantic-relation learning.
5	Experiments	Evaluates SQuAD models under AddSent and other adversarial evaluations; proposes AddSentDiverse and semantic-relation improvements.
6	Results	Reports a 36.5% F1 increase under several adversarial evaluations while maintaining regular SQuAD performance.
7	Limitations	Benefits depend on attack type and generated-example quality.
8	Conclusion	For this project, the defensible takeaway is: Use as a QA-specific precedent for adversarial training. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use as a QA-specific precedent for adversarial training.

20. Beyond Accuracy: Behavioral Testing of NLP Models with CheckList

Authors: Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, Sameer Singh
Publication: ACL, 2020
Research area: Robustness & evaluation
Primary source: https://arxiv.org/abs/2005.04118

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces capability matrices, test types and templates for behavioural testing. Evidence/result reported in the paper: Behavioural tests uncover errors missed by aggregate metrics.
2	Introduction	The central problem is requires task-specific test design. The paper studies this within robustness & evaluation.
3	Related Work	This work is most closely connected to robustness & evaluation and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces capability matrices, test types and templates for behavioural testing.
5	Experiments	Demonstrates targeted tests across NLP systems and tasks.
6	Results	Behavioural tests uncover errors missed by aggregate metrics.
7	Limitations	Requires task-specific test design; it is not a model architecture.
8	Conclusion	For this project, the defensible takeaway is: Create tests for negation, paraphrase, entity changes, distractors and answerability. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Create tests for negation, paraphrase, entity changes, distractors and answerability.

21. Show Your Work: Improved Reporting of Experimental Results

Authors: Jesse Dodge, Suchin Gururangan, Dallas Card, Roy Schwartz, Noah A. Smith
Publication: EMNLP-IJCNLP, 2019
Research area: Evaluation & reproducibility
Primary source: https://arxiv.org/abs/1909.03004

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Recommends transparent reporting of experimental details, variability and tuning decisions. Evidence/result reported in the paper: Supports more reliable and comparable experimental claims.

No.	Section	Structured analysis
2	Introduction	The central problem is does not propose a qa method or directly address data scarcity. The paper studies this within evaluation & reproducibility.
3	Related Work	This work is most closely connected to evaluation & reproducibility and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Recommends transparent reporting of experimental details, variability and tuning decisions.
5	Experiments	Analyzes reporting practices and reproducibility issues in NLP.
6	Results	Supports more reliable and comparable experimental claims.
7	Limitations	Does not propose a QA method or directly address data scarcity.
8	Conclusion	For this project, the defensible takeaway is: Report seeds, tuning budgets, data splits, mean scores and standard deviations. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Report seeds, tuning budgets, data splits, mean scores and standard deviations.

22. Energy and Policy Considerations for Deep Learning in NLP

Authors: Emma Strubell, Ananya Ganesh, Andrew McCallum
Publication: ACL, 2019
Research area: Evaluation & efficiency
Primary source: https://arxiv.org/abs/1906.02243

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Estimates energy, financial and environmental costs of deep-learning NLP training. Evidence/result reported in the paper: Shows model selection should consider compute and energy as well as accuracy.
2	Introduction	The central problem is estimates depend on hardware and energy assumptions. The paper studies this within evaluation & efficiency.
3	Related Work	This work is most closely connected to evaluation & efficiency and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Estimates energy, financial and environmental costs of deep-learning NLP training.
5	Experiments	Compares energy demands across selected NLP training configurations.
6	Results	Shows model selection should consider compute and energy as well as accuracy.
7	Limitations	Estimates depend on hardware and energy assumptions; not a QA benchmark.
8	Conclusion	For this project, the defensible takeaway is: Report parameter count, training time, inference latency and hardware. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Report parameter count, training time, inference latency and hardware.

23. Exploring the Landscape of Distributional Robustness for Question Answering Models

Authors: Anas Awadalla, Mitchell Wortsman, Gabriel Ilharco, Sewon Min, Ian Magnusson, Hannaneh Hajishirzi, Ludwig Schmidt
Publication: Findings of EMNLP, 2022
Research area: Recent robustness & generalization
Primary source: https://aclanthology.org/2022.findings-emnlp.441/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Performs broad empirical comparisons of QA models and adaptation strategies under distribution shifts. Evidence/result reported in the paper: Finds in many cases in-distribution performance predicts out-of-distribution performance; few-shot prompt-tuned models can be more robust than few-shot span-prediction models.
2	Introduction	The central problem is not a controlled technical-domain comparison of the five planned bert variants. The paper studies this within recent robustness & generalization.
3	Related Work	This work is most closely connected to recent robustness & generalization and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.

No.	Section	Structured analysis
4	Methodology	Performs broad empirical comparisons of QA models and adaptation strategies under distribution shifts.
5	Experiments	Evaluates more than 350 models across 16 QA datasets, spanning architectures, model sizes and adaptation methods.
6	Results	Finds in many cases in-distribution performance predicts out-of-distribution performance; few-shot prompt-tuned models can be more robust than few-shot span-prediction models.
7	Limitations	Not a controlled technical-domain comparison of the five planned BERT variants.
8	Conclusion	For this project, the defensible takeaway is: Evaluate all candidate models on held-out domains. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Evaluate all candidate models on held-out domains.

24. Robust Question Answering against Distribution Shifts with Test-Time Adaptation: An Empirical Study

Authors: Hai Ye, Yuyang Ding, Juntao Li, Hwee Tou Ng
Publication: Findings of EMNLP, 2022
Research area: Recent robustness & domain shift
Primary source: https://aclanthology.org/2022.findings-emnlp.460/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Studies online/test-time adaptation for QA under input and domain shifts. Evidence/result reported in the paper: Finds test-time adaptation is comparable to robustness tuning, and applying it after robustness tuning can further improve COLDQA performance.
2	Introduction	The central problem is focuses on xlm-r and test-time adaptation, not limited-data bert comparison. The paper studies this within recent robustness & domain shift.
3	Related Work	This work is most closely connected to recent robustness & domain shift and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Studies online/test-time adaptation for QA under input and domain shifts.
5	Experiments	Introduces COLDQA, covering text corruption and language/domain shifts; evaluates test-time adaptation against robustness tuning.
6	Results	Finds test-time adaptation is comparable to robustness tuning, and applying it after robustness tuning can further improve COLDQA performance.
7	Limitations	Focuses on XLM-R and test-time adaptation, not limited-data BERT comparison.
8	Conclusion	For this project, the defensible takeaway is: Keep domain shift and adversarial perturbation as separate test conditions. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Keep domain shift and adversarial perturbation as separate test conditions.

25. The Impacts of Unanswerable Questions on the Robustness of Machine Reading Comprehension Models

Authors: Son Quoc Tran, Phong Nguyen-Thuan Do, Uyen Le, Matt Kretchmar
Publication: EACL, 2023
Research area: Recent adversarial robustness
Primary source: https://aclanthology.org/2023.eacl-main.113/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Studies how training with unanswerable questions affects MRC robustness. Evidence/result reported in the paper: SQuAD 2.0 training does not immediately make models more robust, but exposes hidden robustness that can be leveraged for gains and transfers to additional OOD datasets.
2	Introduction	The central problem is does not provide the full controlled architecture comparison proposed here. The paper studies this within recent adversarial robustness.
3	Related Work	This work is most closely connected to recent adversarial robustness and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Studies how training with unanswerable questions affects MRC robustness.

No.	Section	Structured analysis
5	Experiments	Fine-tunes three language models on SQuAD 1.1 or SQuAD 2.0 and evaluates robustness under adversarial attacks and out-of-domain data.
6	Results	SQuAD 2.0 training does not immediately make models more robust, but exposes hidden robustness that can be leveraged for gains and transfers to additional OOD datasets.
7	Limitations	Does not provide the full controlled architecture comparison proposed here.
8	Conclusion	For this project, the defensible takeaway is: Include answerability/abstention tests where the chosen dataset supports them. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Include answerability/abstention tests where the chosen dataset supports them.

26. To Adapt or to Annotate: Challenges and Interventions for Domain Adaptation in Open-Domain Question Answering

Authors: Dheeru Dua, Emma Strubell, Sameer Singh, Pat Verga
Publication: ACL, 2023
Research area: Recent domain adaptation
Primary source: https://aclanthology.org/2023.acl-long.807/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Compares interventions for adapting end-to-end open-domain QA systems to target domains. Evidence/result reported in the paper: Shows end-to-end systems fail to generalize; tested interventions improve performance by up to approximately 24 points depending on shift.
2	Introduction	The central problem is open-domain retrieval plus reader differs from an isolated extractive qa reader comparison. The paper studies this within recent domain adaptation.
3	Related Work	This work is most closely connected to recent domain adaptation and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Compares interventions for adapting end-to-end open-domain QA systems to target domains.
5	Experiments	Evaluates end-to-end ODQA under domain shift across five diverse domains and tests data-augmentation interventions.
6	Results	Shows end-to-end systems fail to generalize; tested interventions improve performance by up to approximately 24 points depending on shift.
7	Limitations	Open-domain retrieval plus reader differs from an isolated extractive QA reader comparison.
8	Conclusion	For this project, the defensible takeaway is: Use its domain-shift framing; keep the proposed task extractive and technical. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use its domain-shift framing; keep the proposed task extractive and technical.

27. Generative Data Augmentation using LLMs improves Distributional Robustness in Question Answering

Authors: Arijit Chowdhury, Aman Chadha
Publication: EACL Student Research Workshop, 2024
Research area: Recent domain adaptation & augmentation
Primary source: https://aclanthology.org/2024.eacl-srw.20/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Uses a two-stage LLM generation approach to create contexts and QA pairs for augmentation. Evidence/result reported in the paper: Reports improved robustness to natural distribution shifts after augmenting reading-comprehension datasets with generated contexts and QA pairs.
2	Introduction	The central problem is does not establish equal benefit across bert variants or adversarial attacks. The paper studies this within recent domain adaptation & augmentation.
3	Related Work	This work is most closely connected to recent domain adaptation & augmentation and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Uses a two-stage LLM generation approach to create contexts and QA pairs for augmentation.
5	Experiments	Evaluates generated-data augmentation on four QA datasets under varying degrees of natural distribution shift.

No.	Section	Structured analysis
6	Results	Reports improved robustness to natural distribution shifts after augmenting reading-comprehension datasets with generated contexts and QA pairs.
7	Limitations	Does not establish equal benefit across BERT variants or adversarial attacks.
8	Conclusion	For this project, the defensible takeaway is: Test whether generated technical QA data helps models under low annotation budgets. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Test whether generated technical QA data helps models under low annotation budgets.

28. Towards Robust Extractive Question Answering Models: Rethinking the Training Methodology

Authors: Son Quoc Tran, Matt Kretchmar
Publication: Findings of EMNLP, 2024
Research area: Recent adversarial robustness
Primary source: https://aclanthology.org/2024.findings-emnlp.121/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces a modified training objective/methodology for extractive QA robustness. Evidence/result reported in the paper: Reports an overall F1 improvement of 5.7 across test sets and an adversarial performance decrease of about one-third of the default models' decrease.
2	Introduction	The central problem is focuses on training methodology, not the five-model comparative study. The paper studies this within recent adversarial robustness.
3	Related Work	This work is most closely connected to recent adversarial robustness and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces a modified training objective/methodology for extractive QA robustness.
5	Experiments	Evaluates a new extractive-QA loss/training method on in-domain, out-of-domain and two adversarial conditions.
6	Results	Reports an overall F1 improvement of 5.7 across test sets and an adversarial performance decrease of about one-third of the default models' decrease.
7	Limitations	Focuses on training methodology, not the five-model comparative study.
8	Conclusion	For this project, the defensible takeaway is: Use as a QA-specific robustness training baseline. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use as a QA-specific robustness training baseline.

29. Exploring Language Model Generalization in Low-Resource Extractive QA

Authors: Saptarshi Sengupta, Wenpeng Yin, Preslav Nakov, Shreya Ghosh, Suhang Wang
Publication: COLING, 2025
Research area: Recent domain generalization & data scarcity
Primary source: https://aclanthology.org/2025.coling-main.474/

Source note: Verify final proceedings identifier and exact experimental details.

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Empirically studies zero-shot generalization in low-resource closed-domain extractive QA. Evidence/result reported in the paper: Finds that scaling parameters does not always solve cross-domain generalization and that LLMs struggle with some closed-domain EQA requirements.
2	Introduction	The central problem is zero-shot llm evaluation is not limited-data fine-tuning of bert variants. The paper studies this within recent domain generalization & data scarcity.
3	Related Work	This work is most closely connected to recent domain generalization & data scarcity and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Empirically studies zero-shot generalization in low-resource closed-domain extractive QA.
5	Experiments	Studies zero-shot extractive QA under domain drift, including closed-domain medicine and law, and analyzes answer-span length, domain-specific word senses and dataset differences.
6	Results	Finds that scaling parameters does not always solve cross-domain generalization and that LLMs struggle with some closed-domain EQA requirements.

No.	Section	Structured analysis
7	Limitations	Zero-shot LLM evaluation is not limited-data fine-tuning of BERT variants.
8	Conclusion	For this project, the defensible takeaway is: Use to motivate technical-domain evaluation, but classify as supporting rather than direct BERT scarcity evidence. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use to motivate technical-domain evaluation, but classify as supporting rather than direct BERT scarcity evidence.

30. Fine-tuning Strategies for Domain Specific Question Answering under Low Annotation Budget Constraints

Authors: Kunpeng Guo, Dennis Diefenbach, Antoine Gourru, Christophe Gravier
Publication: IEEE ICTAI 2023, 2023
Research area: Data scarcity
Primary source: https://ieeexplore.ieee.org/document/10356566/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Compares target-only fine-tuning with approaches that combine target-domain QA data and SQuAD. Evidence/result reported in the paper: The authors report that fine-tuning on a combined target-domain plus SQuAD dataset beats the standard strategy by 2.28%–6.48% without extra annotation.
2	Introduction	The central problem is does not compare all planned bert variants across domain adaptation, scarcity and adversarial robustness. The paper studies this within data scarcity.
3	Related Work	This work is most closely connected to data scarcity and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Compares target-only fine-tuning with approaches that combine target-domain QA data and SQuAD.
5	Experiments	Compares alternative sequential fine-tuning strategies across QA datasets with low target-domain annotation budgets.
6	Results	The authors report that fine-tuning on a combined target-domain plus SQuAD dataset beats the standard strategy by 2.28%–6.48% without extra annotation.
7	Limitations	Does not compare all planned BERT variants across domain adaptation, scarcity and adversarial robustness.
8	Conclusion	For this project, the defensible takeaway is: Run technical-only versus SQuAD-plus-technical fine-tuning at fixed low-data budgets. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Run technical-only versus SQuAD-plus-technical fine-tuning at fixed low-data budgets.

31. Few-shot Unified Question Answering: Tuning Models or Prompts?

Authors: Srijan Bansal, Semih Yavuz, Bo Pang, Meghana Bhat, Yingbo Zhou
Publication: Findings of EMNLP, 2023
Research area: Data scarcity
Primary source: https://aclanthology.org/2023.findings-emnlp.550/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Compares full model tuning and prompt tuning, including prompt initialization and parameter sharing. Evidence/result reported in the paper: Prompt tuning can match or outperform model tuning in few-shot settings with good initialization.
2	Introduction	The central problem is unified qa/t5-oriented setting, not a direct five-encoder bert extractive qa comparison. The paper studies this within data scarcity.
3	Related Work	This work is most closely connected to data scarcity and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Compares full model tuning and prompt tuning, including prompt initialization and parameter sharing.
5	Experiments	Evaluates 16 QA datasets across domains and reasoning skills.
6	Results	Prompt tuning can match or outperform model tuning in few-shot settings with good initialization.
7	Limitations	Unified QA/T5-oriented setting, not a direct five-encoder BERT extractive QA comparison.
8	Conclusion	For this project, the defensible takeaway is: Consider parameter-efficient tuning as a separate low-resource baseline. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Consider parameter-efficient tuning as a separate low-resource baseline.

32. QAmeleon: Multilingual QA with Only 5 Examples

Authors: Priyanka Agrawal, Chris Alberti, Fantine Huot, Joshua Maynez, Ji Ma, Sebastian Ruder, Kuzman Ganchev, Dipanjan Das, Mirella Lapata
Publication: TACL, 2023
Research area: Data scarcity
Primary source: https://aclanthology.org/2023.tacl-1.98/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Prompt-tunes a pretrained model to synthesize multilingual QA data from five examples per language. Evidence/result reported in the paper: Outperforms translation-based baselines and closes a substantial part of the gap to a fully supervised upper bound in reported multilingual settings.
2	Introduction	The central problem is focuses on multilingual qa rather than technical-domain bert architecture comparison. The paper studies this within data scarcity.
3	Related Work	This work is most closely connected to data scarcity and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Prompt-tunes a pretrained model to synthesize multilingual QA data from five examples per language.
5	Experiments	Evaluates on TyDiQA-GoldP and MLQA.
6	Results	Outperforms translation-based baselines and closes a substantial part of the gap to a fully supervised upper bound in reported multilingual settings.
7	Limitations	Focuses on multilingual QA rather than technical-domain BERT architecture comparison.
8	Conclusion	For this project, the defensible takeaway is: Use as evidence that a few examples can seed synthetic QA data generation. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Use as evidence that a few examples can seed synthetic QA data generation.

33. Dealing with Data Scarcity in Spoken Question Answering

Authors: Merve Ünlü Menev■e, Yusufcan Manav, Ebru Arisoy, Arzucan Özgür
Publication: LREC-COLING, 2024
Research area: Data scarcity (supporting)
Primary source: https://aclanthology.org/2024.lrec-main.397/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Trains a QA generation model on annotated data, generates QA pairs from unannotated paragraphs and combines them through a selected fine-tuning strategy. Evidence/result reported in the paper: Reports a 5.5% relative F1 gain over training on annotated data alone in the described setting.
2	Introduction	The central problem is spoken input and asr noise differ from text-based technical extractive qa. The paper studies this within data scarcity (supporting).
3	Related Work	This work is most closely connected to data scarcity (supporting) and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Trains a QA generation model on annotated data, generates QA pairs from unannotated paragraphs and combines them through a selected fine-tuning strategy.
5	Experiments	Evaluates spoken QA under limited annotation and varying ASR-error conditions.
6	Results	Reports a 5.5% relative F1 gain over training on annotated data alone in the described setting.
7	Limitations	Spoken input and ASR noise differ from text-based technical extractive QA.
8	Conclusion	For this project, the defensible takeaway is: Adapt the idea of generating QA pairs from unannotated technical documents. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Adapt the idea of generating QA pairs from unannotated technical documents.

34. PPDAC: A Plug-and-Play Data Augmentation Component for Few-shot Extractive Question Answering

Authors: Qi Huang, Han Fu, Wenbin Luo, Mingwen Wang, Kaiwei Luo
Publication: CCL 2024, 2024

Research area: Data scarcity
Primary source: https://aclanthology.org/2024.ccl-1.102/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Uses a paraphrase generator and paraphrase selector to create additional questions for (question, passage, answer) triples. Evidence/result reported in the paper: Reports that selected question paraphrases improve few-shot QA performance.
2	Introduction	The central problem is paraphrases may change meaning or answerability. The paper studies this within data scarcity.
3	Related Work	This work is most closely connected to data scarcity and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Uses a paraphrase generator and paraphrase selector to create additional questions for (question, passage, answer) triples.
5	Experiments	Evaluates few-shot extractive QA across several datasets, including SQuAD, NewsQA, TriviaQA, HotpotQA, Natural Questions, BioASQ and TextbookQA.
6	Results	Reports that selected question paraphrases improve few-shot QA performance.
7	Limitations	Paraphrases may change meaning or answerability; augmentation needs quality checks.
8	Conclusion	For this project, the defensible takeaway is: Compare original limited data with paraphrase-augmented data for each encoder. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Compare original limited data with paraphrase-augmented data for each encoder.

35. Improving Low-resource Question Answering by Augmenting Question Information

Authors: Andong Chen, Yuan Sun, Xiaobing Zhao, Rosella P. Galindo Esparza, Kehai Chen, Yang Xiang, Tiejun Zhao, Min Zhang
Publication: Findings of EMNLP, 2023
Research area: Data scarcity
Primary source: https://aclanthology.org/2023.findings-emnlp.699/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces PQQ with Prompt Answer, Question Generation and Question Filter stages. Evidence/result reported in the paper: Reports that PQQ outperforms compared augmentation strategies in the studied settings.
2	Introduction	The central problem is depends on generated-question quality and filtering. The paper studies this within data scarcity.
3	Related Work	This work is most closely connected to data scarcity and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Introduces PQQ with Prompt Answer, Question Generation and Question Filter stages.
5	Experiments	Evaluates on PolicyQA and TechQA, plus SQuAD 1.1 and TriviaQA.
6	Results	Reports that PQQ outperforms compared augmentation strategies in the studied settings.
7	Limitations	Depends on generated-question quality and filtering; not a BERT-variant robustness comparison.
8	Conclusion	For this project, the defensible takeaway is: Especially relevant because TechQA is a technical-domain QA dataset. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Especially relevant because TechQA is a technical-domain QA dataset.

36. Prompting-based Synthetic Data Generation for Few-Shot Question Answering

Authors: Maximilian Schmidt, Andrea Bartezzaghi, Ngoc Thang Vu
Publication: LREC-COLING, 2024
Research area: Data scarcity
Primary source: https://aclanthology.org/2024.lrec-main.1153/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Uses prompting to generate synthetic QA examples from documents, then trains QA models with generated examples. Evidence/result reported in the paper: Reports consistent improvements over previous few-shot approaches in evaluated settings.

No.	Section	Structured analysis
2	Introduction	The central problem is synthetic quality and domain relevance remain concerns. The paper studies this within data scarcity.
3	Related Work	This work is most closely connected to data scarcity and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Uses prompting to generate synthetic QA examples from documents, then trains QA models with generated examples.
5	Experiments	Evaluates few-shot extractive QA across multiple MRQA datasets and shot counts.
6	Results	Reports consistent improvements over previous few-shot approaches in evaluated settings.
7	Limitations	Synthetic quality and domain relevance remain concerns; it does not compare all proposed BERT variants.
8	Conclusion	For this project, the defensible takeaway is: Compare limited original data with limited-plus-synthetic technical QA data. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Compare limited original data with limited-plus-synthetic technical QA data.

37. CoDa: Constrained Generation based Data Augmentation for Low-Resource NLP

Authors: Chandra Kiran Evuru, Sreyan Ghosh, Sonal Kumar, Ramaneswaran S, Utkarsh Tyagi, Dinesh Manocha
Publication: Findings of NAACL, 2024
Research area: Data scarcity (supporting)
Primary source: https://aclanthology.org/2024.findings-naacl.238/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Extracts constraints from low-resource examples and prompts an instruction-following LLM to generate examples satisfying them. Evidence/result reported in the paper: Reports improvements of 0.12%–7.19% across evaluated tasks/settings.
2	Introduction	The central problem is not exclusively extractive qa. The paper studies this within data scarcity (supporting).
3	Related Work	This work is most closely connected to data scarcity (supporting) and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.
4	Methodology	Extracts constraints from low-resource examples and prompts an instruction-following LLM to generate examples satisfying them.
5	Experiments	Evaluates 11 datasets, three NLP tasks and three low-resource settings.
6	Results	Reports improvements of 0.12%–7.19% across evaluated tasks/settings.
7	Limitations	Not exclusively extractive QA; generated examples require task-specific validation.
8	Conclusion	For this project, the defensible takeaway is: Constrain generated technical QA examples to preserve terminology, answer types and consistency. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Constrain generated technical QA examples to preserve terminology, answer types and consistency.

38. Few-Shot Data Synthesis for Open Domain Multi-Hop Question Answering

Authors: Mingda Chen, Xilun Chen, Wen-tau Yih
Publication: EACL, 2024
Research area: Data scarcity (supporting)
Primary source: https://aclanthology.org/2024.eacl-long.12/

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Uses document relationships and LLM prompts to synthesize multi-hop questions, claims and search queries from fewer than ten annotated QA pairs. Evidence/result reported in the paper: Reports improvements for smaller models and competitive results against larger-model approaches in the studied tasks.
2	Introduction	The central problem is open-domain multi-hop qa differs from closed-context technical extractive qa. The paper studies this within data scarcity (supporting).
3	Related Work	This work is most closely connected to data scarcity (supporting) and should be read alongside the project's other papers on domain transfer, low-resource supervision and QA evaluation. Its task and benchmark are not automatically interchangeable with technical-domain extractive QA.

No.	Section	Structured analysis
4	Methodology	Uses document relationships and LLM prompts to synthesize multi-hop questions, claims and search queries from fewer than ten annotated QA pairs.
5	Experiments	Evaluates HotpotQA, 2WikiMultiHopQA, MuSiQue and FEVER.
6	Results	Reports improvements for smaller models and competitive results against larger-model approaches in the studied tasks.
7	Limitations	Open-domain multi-hop QA differs from closed-context technical extractive QA.
8	Conclusion	For this project, the defensible takeaway is: Consider generating linked questions from related technical documents only if the task is expanded to multi-hop QA. The result should be treated as evidence for the paper's own setting, not assumed to transfer unchanged.

How it helps our project: Consider generating linked questions from related technical documents only if the task is expanded to multi-hop QA.

39. Improving model adversarial robustness in Extractive Question Answering via Wasserstein-Guided feature Representations

Authors: Gang Huang, Lu Zhang, Hailun Wang
Publication: Alexandria Engineering Journal, 2025
Research area: Recent adversarial robustness
Primary source: https://doi.org/10.1016/j.aej.2025.05.003

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Proposes Wasserstein-Distance-Guided feature Representations (WDGR), an adversarial training method that operates on perturbed feature representations rather than generating complete adversarial passages. Evidence/result reported in the paper: Publisher abstract motivates WDGR as a way to reduce the computational expense and noise associated with full adversarial-passage generation.
2	Introduction	The paper addresses a challenge in question answering or language-model development relevant to this research dimension.
3	Related Work	The work builds on pretrained language models, transfer learning and prior question-answering or NLP evaluation methods.
4	Methodology	Proposes Wasserstein-Distance-Guided feature Representations (WDGR), an adversarial training method that operates on perturbed feature representations rather than generating complete adversarial passages.
5	Experiments	Studies extractive QA robustness against adversarial perturbations and compares the proposed feature-space approach with existing adversarial training strategies.
6	Results	Publisher abstract motivates WDGR as a way to reduce the computational expense and noise associated with full adversarial-passage generation.
7	Limitations	Not specifically a technical-domain, low-annotation BERT-variant comparison; full-text verification is required for detailed metrics and tested model configurations.
8	Conclusion	The paper provides evidence for its stated task and experimental setting. Transfer to the proposed technical extractive-QA setting must be tested rather than assumed.

How it helps our project: Include as a recent EQA robustness method and compare its motivation with token-level attacks and embedding-level adversarial training.

40. CATS-RAG: Contextual Augmented Triplet Synthesis for RAG in Technical QA

Authors: Changwook Chu, Yongtae Jeong, Hansam Cho, Jaehoon Kim, Jungmin Lee, Byungwoo Bang, Junyeon Lee, Uiseok Song, Seoung Bum Kim
Publication: Expert Systems with Applications, 2026
Research area: Recent technical QA & data scarcity
Primary source: https://doi.org/10.1016/j.eswa.2026.131491

Source note: Publisher page and abstract verified; final page-level details should be checked in the full article.

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces a contextual augmented triplet synthesis framework for retrieval-augmented generation in technical QA, generating document-question-answer triplets and fine-tuning retriever and generator components. Evidence/result reported in the paper: Reports average retriever improvements of 4.3% on TechQA and 6.5% on Microsoft QA; generator improvements of 1.7% and 11.9%, respectively.
2	Introduction	The paper addresses a challenge in question answering or language-model development relevant to this research dimension.

No.	Section	Structured analysis
3	Related Work	The work builds on pretrained language models, transfer learning and prior question-answering or NLP evaluation methods.
4	Methodology	Introduces a contextual augmented triplet synthesis framework for retrieval-augmented generation in technical QA, generating document-question-answer triplets and fine-tuning retriever and generator components.
5	Experiments	Evaluates on TechQA and Microsoft QA, using generated triplets and hard distractor documents.
6	Results	Reports average retriever improvements of 4.3% on TechQA and 6.5% on Microsoft QA; generator improvements of 1.7% and 11.9%, respectively.
7	Limitations	RAG pipeline rather than extractive BERT span prediction; does not directly compare BERT-family encoders under matched low-data and adversarial conditions.
8	Conclusion	The paper provides evidence for its stated task and experimental setting. Transfer to the proposed technical extractive-QA setting must be tested rather than assumed.

How it helps our project: Important adjacent work: it shows technical QA data scarcity and robustness are active research areas, so our gap must focus on extractive encoder comparison and controlled interactions.

41. Toward construction-specialized, small language models: The interplay of domain adaptation, model scale and data volume

Authors: Shuyi Wang, Yuguang Fu, Jinwoo Kim
Publication: Advanced Engineering Informatics, 2026
Research area: Recent technical QA & domain adaptation
Primary source: https://doi.org/10.1016/j.aei.2025.104035

Source note: Publisher metadata and abstract verified; confirm full metric definition and experimental details from the full article.

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Studies small, medium and large construction-specialized language models and compares prompt engineering, RAG, task-specific fine-tuning, and pretraining-plus-fine-tuning. Evidence/result reported in the paper: Publisher abstract reports that a small model adapted through pretraining-plus-fine-tuning achieved the strongest result, with a 14.6% F1 improvement and fourfold inference speed over larger counterparts.
2	Introduction	The paper addresses a challenge in question answering or language-model development relevant to this research dimension.
3	Related Work	The work builds on pretrained language models, transfer learning and prior question-answering or NLP evaluation methods.
4	Methodology	Studies small, medium and large construction-specialized language models and compares prompt engineering, RAG, task-specific fine-tuning, and pretraining-plus-fine-tuning.
5	Experiments	Evaluates on a construction-specific QA dataset and compares domain-adaptation strategies across model scales.
6	Results	Publisher abstract reports that a small model adapted through pretraining-plus-fine-tuning achieved the strongest result, with a 14.6% F1 improvement and fourfold inference speed over larger counterparts.
7	Limitations	Construction-domain language models and QA may use generative setups rather than the proposed extractive BERT span-prediction comparison.
8	Conclusion	The paper provides evidence for its stated task and experimental setting. Transfer to the proposed technical extractive-QA setting must be tested rather than assumed.

How it helps our project: Shows that model scale, data volume and domain adaptation interact in technical QA; directly relevant to the project's experimental design.

42. Autogenerating a Domain-Specific Question-Answering Data Set to Enable High-Performing Language Models for Magnetic Materials

Authors: Charles R. Kelly, Jacqueline M. Cole
Publication: Journal of Chemical Information and Modeling, 2026
Research area: Recent technical QA & data scarcity
Primary source: https://doi.org/10.1021/acs.jcim.6c00136

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Autogenerates 168,080 magnetic-materials QA pairs (112,043 answerable and 56,037 unanswerable). Compares vanilla BERT-base-cased and DAPT-adapted MagBERT fine-tuned on MagQA, SQuAD v2 or their mixture; also extends the selected setup to RoBERTa-base and DeBERTa-base. Evidence/result reported in the paper: The best reported model is vanilla BERT-base-cased fine-tuned on MagQA plus SQuAD v2, with 78.43 F1 and 72.84 EM. The authors conclude that sufficiently large, high-quality domain QA data can make DAPT unnecessary in their setting.
2	Introduction	The paper addresses a challenge in question answering or language-model development relevant to this research dimension.
3	Related Work	The work builds on pretrained language models, transfer learning and prior question-answering or NLP evaluation methods.
4	Methodology	Autogenerates 168,080 magnetic-materials QA pairs (112,043 answerable and 56,037 unanswerable). Compares vanilla BERT-base-cased and DAPT-adapted MagBERT fine-tuned on MagQA, SQuAD v2 or their mixture; also extends the selected setup to RoBERTa-base and DeBERTa-base.
5	Experiments	Uses a corpus of magnetic-materials papers for DAPT, fine-tunes with QA-data-size ablations (10,000, 50,000, 100,000 and full MagQA), and evaluates on a manually curated held-out magnetic-materials QA set.
6	Results	The best reported model is vanilla BERT-base-cased fine-tuned on MagQA plus SQuAD v2, with 78.43 F1 and 72.84 EM. The authors conclude that sufficiently large, high-quality domain QA data can make DAPT unnecessary in their setting.
7	Limitations	This is the closest prior work: it already studies BERT, RoBERTa and DeBERTa, DAPT versus no-DAPT, and QA-data-size ablations in a scientific domain. Its central contribution is QA-data generation and domain adaptation; the proposed project must verify whether and how adversarial attacks are evaluated before claiming that dimension as absent.
8	Conclusion	This study substantially narrows the proposed gap. The project must build on it rather than present domain-specific BERT comparison and data-volume ablation as new.

How it helps our project: Treat this as the closest comparator. The project cannot claim novelty for merely comparing BERT/RoBERTa/DeBERTa, DAPT versus no-DAPT, or data-size ablations in a technical/scientific QA domain. A defensible distinction is a controlled adversarial evaluation and analysis of the interaction between annotation budget, adaptation and robustness in the chosen technical corpus.

43. Adversarial Question Answering Robustness: A Multi-Level Error Analysis and Mitigation Study

Authors: Agniv Roy Choudhury, Vignesh Ponselvan Rajasingh
Publication: arXiv preprint, 2026
Research area: Recent adversarial robustness
Primary source: https://arxiv.org/abs/2601.02700

Source note: arXiv record and abstract verified; treat as non-peer-reviewed preprint.

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Studies AddSent adversarial robustness in transformer QA models and tests mitigation strategies including adversarial fine-tuning and entity-aware contrastive learning. Evidence/result reported in the paper: The preprint reports entity-aware contrastive learning as its strongest mitigation and reports 89.89% AddSent EM and 90.73% SQuAD EM for its selected configuration.
2	Introduction	The paper addresses a challenge in question answering or language-model development relevant to this research dimension.
3	Related Work	The work builds on pretrained language models, transfer learning and prior question-answering or NLP evaluation methods.
4	Methodology	Studies AddSent adversarial robustness in transformer QA models and tests mitigation strategies including adversarial fine-tuning and entity-aware contrastive learning.
5	Experiments	Compares model scales and adversarial-training ratios and performs multi-level error analysis of failure types.
6	Results	The preprint reports entity-aware contrastive learning as its strongest mitigation and reports 89.89% AddSent EM and 90.73% SQuAD EM for its selected configuration.
7	Limitations	Preprint, not peer-reviewed at the time of this review; reported results require independent reproduction and should not be treated as established consensus.
8	Conclusion	The paper provides evidence for its stated task and experimental setting. Transfer to the proposed technical extractive-QA setting must be tested rather than assumed.

How it helps our project: Very recent adjacent work; use it to ensure the project differentiates its contribution from AddSent mitigation and error-analysis studies.

44. Autogenerating a Domain-Specific Question-Answering Data Set from a Thermoelectric Materials Database to Enable High-Performing BERT Models

Authors: Odysseas Sierepeklis, Jacqueline M. Cole
Publication: Journal of Chemical Information and Modeling, 65(16), 8579–8592, 2025
Research area: Recent technical QA & data scarcity
Primary source: https://doi.org/10.1021/acs.jcim.5c00840

Source note: Publisher record, authors, venue, date and abstract verified from ACS/PubMed.

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Automatically generates a thermoelectric-materials QA dataset of 99,757 pairs from a structured materials database. Fine-tunes BERT on domain-only data, SQuAD v2, and a mixture of both. Evidence/result reported in the paper: The mixed dataset achieves 67.93 EM and 72.29 F1 on the reported thermoelectric test set, outperforming the domain-only and SQuAD-v2-only variants.
2	Introduction	The paper addresses a challenge in question answering or language-model development relevant to this research dimension.
3	Related Work	The work builds on pretrained language models, transfer learning and prior question-answering or NLP evaluation methods.
4	Methodology	Automatically generates a thermoelectric-materials QA dataset of 99,757 pairs from a structured materials database. Fine-tunes BERT on domain-only data, SQuAD v2, and a mixture of both.
5	Experiments	Creates a manually annotated test set and evaluates BERT on domain-specific extractive QA; compares the three fine-tuning datasets.
6	Results	The mixed dataset achieves 67.93 EM and 72.29 F1 on the reported thermoelectric test set, outperforming the domain-only and SQuAD-v2-only variants.
7	Limitations	Uses BERT rather than a broad family of encoders and does not make adversarial robustness the central experimental axis. Domain is thermoelectric materials rather than technical documentation.
8	Conclusion	The paper's findings are valid within its own evaluation setting; transfer to technical extractive QA requires direct testing.

How it helps our project: Directly challenges generic novelty claims about generated domain QA data and mixing domain-specific data with SQuAD. Use it to justify why the project's distinguishing contribution must include systematic robustness evaluation.

45. Automatic Generation of a Mechanical Properties Question-Answering Data Set for Language Model Benchmarking: A Comparative Study of BERT, XLNet, and LLaMA Models

Authors: Minglei Zhang, Jacqueline M. Cole
Publication: Journal of Chemical Information and Modeling, 66(7), 3840–3857, 2026
Research area: Recent technical QA & data scarcity
Primary source: https://doi.org/10.1021/acs.jcim.5c02646

Source note: ACS and PubMed/PMC bibliographic records and abstract verified.

No.	Section	Structured analysis
1	Abstract	Problem and contribution: Introduces MechQA, a dataset of 202,068 QA pairs automatically distilled from 125,967 materials-science articles, and fine-tunes extractive BERT-base and XLNet-base plus generative LLaMA-3.1-Instruct. Evidence/result reported in the paper: Reports dataset quality of 83.76% precision, 89.09% recall and 86.34% F1. On validation, BERT reaches 78.03 EM/84.50 F1, XLNet 78.21 EM/84.70 F1, and LLaMA 80.48 EM/86.25 F1.
2	Introduction	The paper addresses a challenge in question answering or language-model development relevant to this research dimension.
3	Related Work	The work builds on pretrained language models, transfer learning and prior question-answering or NLP evaluation methods.
4	Methodology	Introduces MechQA, a dataset of 202,068 QA pairs automatically distilled from 125,967 materials-science articles, and fine-tunes extractive BERT-base and XLNet-base plus generative LLaMA-3.1-Instruct.
5	Experiments	Manually evaluates dataset quality and trains on 181,722 QA pairs with 20,346 validation pairs; compares EM, F1 and expected calibration error.
6	Results	Reports dataset quality of 83.76% precision, 89.09% recall and 86.34% F1. On validation, BERT reaches 78.03 EM/84.50 F1, XLNet 78.21 EM/84.70 F1, and LLaMA 80.48 EM/86.25 F1.
7	Limitations	Materials-science domain and generated large-scale dataset; the published abstract describes fine-tuning comparisons rather than a low-label adversarial robustness factorial study.
8	Conclusion	The paper's findings are valid within its own evaluation setting; transfer to technical extractive QA requires direct testing.

How it helps our project: Very close current technical/scientific extractive-QA precedent. It means model comparison and automatic domain QA dataset creation are already established; differentiate via low-annotation budgets and adversarial robustness under matched protocols.

Final comparative analysis

The literature does not support a broad claim that domain adaptation, low-resource QA or adversarial QA are individually unexplored. The defensible contribution must be framed around the exact task, corpus, label budget, model family and controlled interaction being studied.

Closest prior work	Already established	Remaining distinction for this project
Sierepeklis & Cole (2025), thermoelectric QA	Automatically generates 99,757 domain QA pairs and fine-tunes BERT on domain-only, SQuAD v2 and mixed data; reports EM/F1 on a manually annotated test set.	Do not claim generated domain QA data or mixing with SQuAD is new. Focus on limited label budgets and adversarial robustness in the chosen technical corpus.
Kelly & Cole (2026), MagQA	Generates 168,080 magnetic-materials QA pairs; compares BERT and DAPT/no-DAPT; extends to RoBERTa/DeBERTa and performs data-size ablations.	Do not claim BERT/RoBERTa/DeBERTa comparison, DAPT and QA-data-size ablation in a scientific domain as novel. Verify exact attack coverage in the full article.
Zhang & Cole (2026), MechQA	Creates 202,068 mechanical-property QA pairs from 125,967 articles; compares extractive BERT/XLNet with generative LLaMA and reports dataset quality and model metrics.	Dataset generation and extractive-model evaluation in technical science are established. Differentiate with controlled low-shot and adversarial robustness experiments.
Huang et al. (2025), WDGR	Proposes feature-space Wasserstein-guided adversarial training for extractive QA; evaluates multiple datasets and attacks.	Measure baseline vulnerability and test whether attack degradation changes with encoder, label budget and domain adaptation; treat WDGR as an optional defence comparator.
Tran & Kretchmar (2024)	Proposes a modified extractive-QA loss and reports improvements on OOD datasets and two adversarial attack types.	Test transfer of robustness findings to the selected technical domain and across BERT-family encoders at different label budgets.
Awadalla et al. (2022)	Evaluates 350+ models across 16 QA datasets for distributional robustness.	A narrower, matched-protocol study can add value through controlled technical extractive QA, explicit label-budget curves and adversarial tests.
CATS-RAG (2026)	Synthesizes technical QA triplets and hard distractors for RAG on TechQA and Microsoft QA.	Clearly distinguish extractive span prediction from retrieval-plus-generation. It is adjacent evidence that technical QA data scarcity and distractors are active research problems.
Wang et al. (2026), construction QA	Studies domain adaptation, model scale and data volume on a construction-specific QA task.	Focus on extractive BERT-family encoders and explicit adversarial robustness measurement under controlled low-data conditions.

Project-specific research gap

Proposed gap statement: Recent studies have separately advanced domain-specific QA data generation and adaptation, low-annotation-budget fine-tuning, distribution-shift robustness, and adversarially robust extractive QA. More importantly, 2025–2026 materials-science studies already generate large domain-specific QA datasets and compare BERT-family or other extractive models, including data-size and adaptation choices. Therefore, the project must not claim novelty for domain-specific BERT fine-tuning, QA data generation, DAPT, model comparison or data-volume ablation alone. A plausible, narrower gap is the lack of a clearly established, matched-budget empirical comparison—within the selected technical-documentation extractive-QA corpus—of how encoder choice and continued domain-adaptive pretraining interact with low labelled-data budgets and adversarial robustness. The study would quantify whether model rankings and robustness degradation change as labelled QA examples decrease, under both natural domain shift and controlled adversarial perturbations.

Research question: Under matched training and tuning budgets, how do selected BERT-family encoders differ in clean answer quality, low-resource sample efficiency and adversarial robustness on a held-out technical-domain extractive QA dataset, and how does continued domain-adaptive pretraining affect these outcomes?

Sub-questions: (1) Which encoder retains the best EM/F1 as labelled QA examples decrease? (2) Does continued MLM pretraining on unlabeled technical documents improve low-data performance, and does it affect adversarial degradation? (3) Which attack types cause the largest degradation, and do model rankings change under attack? (4) Which model offers the best joint trade-off among EM/F1, robustness drop and computational cost?

Proposed contribution: A reproducible, controlled benchmark of BERT, RoBERTa and DeBERTa, with DistilBERT/ALBERT as optional efficiency baselines, on one technical extractive-QA task. Evaluate fixed label budgets (for example 1%, 5%, 10%, 25%, 50%, 100%), with and without DAPT; measure clean in-domain and held-out-domain performance; apply QA-valid attacks such as AddSent-style distractors and carefully adapted lexical perturbations; report EM, token F1, answerability where supported, attack success/degradation, mean and standard deviation across seeds, training time, latency and memory.

Critical novelty caution: The MagQA 2026, MechQA 2026 and thermoelectric QA 2025 papers are close prior art. The research gap is a proposed gap based on this reviewed corpus, not proof of global novelty. Before submission, read their complete methods and evaluation sections and conduct a final database search. In particular, do not state that the MagQA 2026 study excludes adversarial evaluation until its complete experimental sections have been inspected.

Recommended experimental matrix

Factor	Suggested levels	Control / reporting
Encoder	BERT, RoBERTa, DeBERTa; optional DistilBERT and ALBERT	Same QA formulation and tuning budget; document checkpoints and tokenizers.
Domain adaptation	No DAPT vs continued MLM on unlabeled technical corpus	No QA test leakage into DAPT corpus; same downstream fine-tuning protocol.
Label budget	1%, 5%, 10%, 25%, 50%, 100% or feasible fixed shot counts	Reproducible sampling; keep the same test set at every budget.
Test condition	Clean in-domain, held-out domain/shift, adversarial	Keep natural shift and adversarial attacks as separate conditions.
Metrics	EM, token F1, answerability, attack success/degradation	Multiple seeds, mean ± SD; include latency, memory and training time.

Final conclusion

The strongest direction is not to claim a new domain-adaptation method or adversarial attack. It is to produce a carefully controlled empirical study of how established BERT-family encoders behave when technical QA labels are scarce, domain adaptation is varied, and natural shifts and adversarial perturbations are introduced. The 2025–2026 materials-science QA papers make this narrower positioning essential.

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